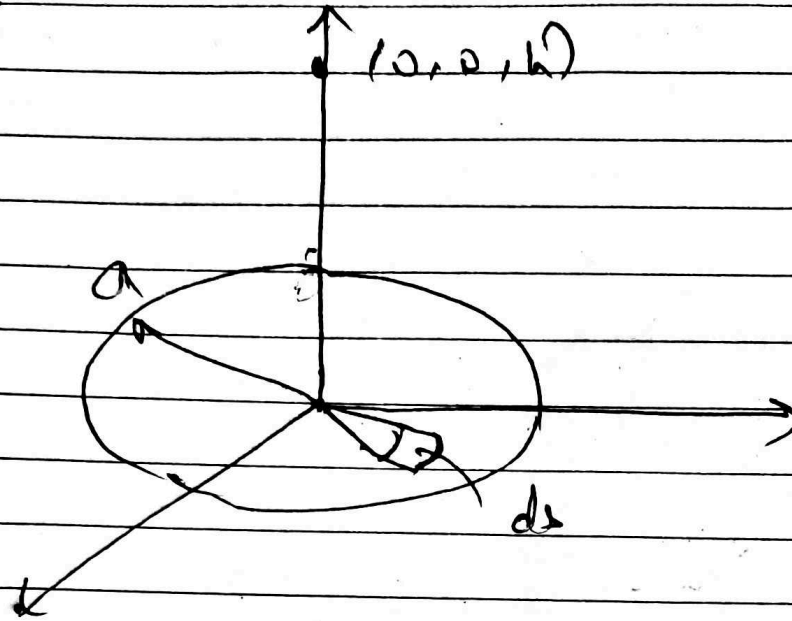


Question 2

Define the vectors.

$$\vec{R}_p = h\hat{z} \quad R_d = r\hat{r}$$

$$\vec{R} = h\hat{z} - r\hat{r}$$

$$|\vec{R}| = (h^2 + r^2)^{1/2}$$

$$\vec{E} = \int \frac{\rho_s ds}{4\pi\epsilon_0} \frac{(h\hat{z} - r\hat{r})}{(h^2 + r^2)^{3/2}} \quad d\vec{s} = r d\phi d\vec{r}$$

$$\vec{E} = \int \frac{\rho_s}{4\pi\epsilon_0} \frac{(h\hat{z} - r\hat{r})}{(h^2 + r^2)^{3/2}} r d\phi d\vec{r}$$

$$\vec{E} = \int \frac{\rho_0 r^4}{4\pi\epsilon_0} \frac{(h\hat{z} - r\hat{r})}{(h^2 + r^2)^{3/2}} r d\phi d\vec{r}$$

$$\vec{E} = \frac{\rho_0}{4\pi\epsilon_0} \int \frac{r^5 (h\hat{z} - r\hat{r})}{(h^2 + r^2)^{3/2}} d\vec{r} d\phi$$

Due to the symmetry all along the  $\hat{r}$  terms since they get cancelled for full circle of  $\phi$  from

0 to  $2\pi$ .

$$\vec{E} = \frac{\rho \cdot \hat{h}}{4\pi\epsilon_0} \int_S \frac{r^5 (\hat{h})}{(h^2 + r^2)^{3/2}}$$

$$\vec{E} = \frac{\rho \cdot \hat{h}}{4\pi\epsilon_0} \int_S \frac{r^5}{(h^2 + r^2)^{3/2}} dr d\phi$$

$$\boxed{u^2 = r^2 + h^2} \quad \text{using substitution.}$$

$$(h^2 + r^2)^{3/2} = (u^2)^{3/2} = u^3$$

$$\frac{d(u^2)}{dr} = \frac{d}{dr} (r^2 + h^2)$$

$$\frac{d(u^2)}{du} \times \frac{du}{dr} = 2r + 0 \quad \text{since } h \text{ is a constant}$$

$$2u \frac{du}{dr} = 2r$$

$$\boxed{u du = r dr}$$

Using the limits now

$r$  varies from 0 to  $a$  for the surface charge density.

At  $r = 0$

$$u^2 = (0)^2 + h^2$$

$$u = h$$

$$\boxed{u = h}$$

Starting point is above the surface charge density.

Question 1 continued

for the upper limit

$$\begin{aligned} u^2 &= r^2 + h^2 \\ u^2 &= a^2 + h^2 \\ u &= \sqrt{a^2 + h^2} \end{aligned}$$

~~$\vec{E} =$~~

$$\vec{E} = \frac{\rho_0}{4\pi\epsilon_0} \int_0^{2\pi} \int_h^{\sqrt{a^2+h^2}} \frac{r^4}{(h^2+r^2)^{3/2}} r dr d\phi$$

Substituting the new variables

$$(h^2+r^2)^{3/2} = (u^2)^{3/2} = u^3$$

$$u^2 = r^2 + h^2$$

$$r^2 = u^2 - h^2$$

$$r^4 = (u^2 - h^2)^2$$

$$\vec{E} = \frac{\rho_0 h^2}{4\pi\epsilon_0} \int_0^{2\pi} \int_h^{\sqrt{a^2+h^2}} \frac{(u^2 - h^2)^2}{u^3} u du d\phi$$

$$I_2 = \frac{\rho_0 h^2}{4\pi \epsilon_0} \int_0^{2\pi} \int_h^{\sqrt{a^2+h^2}} \frac{(u^2 - h^2)^2}{u^2} du d\phi$$

$$(u^2 - h^2)^2 = (u^2)^2 - 2(u^2)(h^2) + (h^2)^2 \\ = u^4 - 2u^2h^2 + h^4$$

$$\frac{(u^2 - h^2)^2}{u^2} = \frac{u^4}{u^2} - \frac{2u^2h^2}{u^2} + \frac{h^4}{u^2} \\ = u^2 - 2h^2 + \frac{h^4}{u^2}$$

$$I_2 = \int_h^{\sqrt{a^2+h^2}} \frac{u^2 - 2h^2 + \frac{h^4}{u^2}}{u^2} du$$

$$= \left[ \frac{u^3}{3} - 2h^2u + \frac{h^4 u^{-2+1}}{-2} \right]_h^{\sqrt{a^2+h^2}}$$

$$= \left[ \frac{(a^2+h^2)^{3/2}}{3} - 2h^2(a^2+h^2)^{1/2} - \frac{h^4}{(a^2+h^2)^{1/2}} \right] - \\ \left( \frac{h^3}{3} - 2h^2(h) - \frac{h^4}{h} \right)$$

$$= \left[ \frac{(a^2+h^2)^{3/2}}{3} - 2h^2(a^2+h^2)^{1/2} - \frac{h^4}{(a^2+h^2)^{1/2}} - \left( \frac{h^3}{3} - 2h^3 - h^3 \right) \right]$$

$$= \left[ \frac{(a^2+h^2)^{3/2}}{3} - 2h^2(a^2+h^2)^{1/2} - \frac{h^4}{(a^2+h^2)^{1/2}} + \frac{8h^3}{3} \right]$$

continued on extra  
sheet

Extra sheet

Zaryan

Instructor  
comments  
do not write  
in this section

Question 1 continued

$$\vec{E} = \frac{\rho_0 h \hat{z}}{4\pi\epsilon_0} (2\pi) \left[ \frac{(a^2 + h^2)^{3/2}}{3} - 2h^2(a^2 + h^2)^{1/2} - h^4 + \frac{8h^3}{3} \right]$$

Therefore the electric field at  $(0, 0, h)$  is

$$\vec{E} = \frac{\rho_0 h}{4\pi\epsilon_0} \left[ \frac{(a^2 + h^2)^{3/2}}{3} - 2h^2(a^2 + h^2)^{1/2} - h^4 + \frac{8h^3}{3} \right] \hat{z} \quad \text{2 Ans}$$

Well done



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